

# A Matrix- and Phenotype-Aware Digital Twin for Cell Migration Under Interstitial Flow

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## Abstract

Interstitial flow can reorganize cancer-cell migration through extracellular-matrix remodeling, altered cell-matrix adhesion, phenotype switching, and stochastic motility. We develop a proof-of-concept digital twin that couples flow-dependent fibronectin and adhesion dynamics to amoeboid-mesenchymal switching, active colored noise, translational fluctuations, excluded volume, and short-range cell-cell adhesion. Published phenotype-resolved observables constrain the mechanistic model, while synthetic microscopy experiments with withheld ground truth test online state estimation and forecasting. In a held-out evaluation of 640 trajectories, simulated mesenchymal persistence, mesenchymal mean-squared-displacement (MSD) exponent, amoeboid MSD exponent, and the amoeboid/mesenchymal speed ratio differed from published targets by 3.3%,  $-1.1\%$ ,  $-4.6\%$ , and  $-4.1\%$ ; amoeboid persistence was 16.2% high. Across ten independent interacting-population experiments, a hybrid ensemble Kalman/particle update reduced position RMSE from  $0.700 \pm 0.009$  for raw measurements to  $0.689 \pm 0.008$  ( $1.63 \pm 0.61\%$ ; paired  $p = 1.57 \times 10^{-5}$ ), with 93.4% coverage for nominal 95% intervals. Latent phenotype recovery was weak (ROC AUC  $0.565 \pm 0.050$ ), showing that coordinates alone were insufficient for reliable frame-level phenotype reconstruction. In ten independent withheld-forecast experiments, the twin reduced RMSE relative to linear extrapolation by 9.9–48.7% over horizons of 0.5–8 model-time units, with 93.4% mean coverage. The model further predicts flow-induced compaction through reduced persistence-driven dispersal; an ablation showed that this effect was not driven by the weak short-range pair force. These results provide a reproducible synthetic-data demonstration while defining clear limits on phenotype identifiability and experimental interpretation.

## 1 Introduction

Metastatic dissemination requires cells to migrate through mechanically and chemically heterogeneous tissues while switching among collective and individual modes [1–3]. In three-dimensional matrices, cells can alternate between elongated, adhesion-dependent mesenchymal migration and rounded, weakly adherent amoeboid migration [4–7]. These modes differ in persistence, speed, force transmission, and matrix engagement, making phenotype a natural hidden state in predictive migration models.

Interstitial flow transports soluble factors, reshapes gradients, and modifies mechanical loading [8–11]. Huang *et al.* reported flow-dependent changes in phenotype fractions and phenotype-resolved motility in a three-dimensional breast-cancer model [12]. A complementary active-matter theory connected flow-dependent adhesion to migration heterogeneity and collective behavior [13]. These observations motivate a model representing the causal chain from flow to matrix state, cell-matrix adhesion, phenotype, and motion rather than fitting trajectories alone.

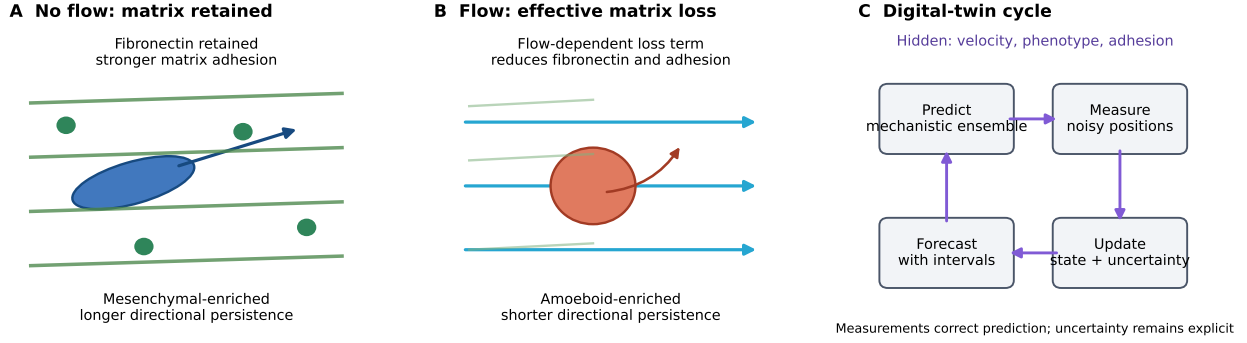


Figure 1: Modeled biological hypothesis and digital-twin workflow. (A) Without flow, retained fibronectin supports stronger effective cell–matrix adhesion and enriches longer-persistence mesenchymal migration. (B) Under flow, an effective matrix-loss term reduces fibronectin and adhesion, enriching shorter-persistence amoeboid migration. Blue arrows indicate the imposed flow condition, not a computed spatial velocity field. (C) Noisy position measurements repeatedly correct an ensemble mechanistic prediction, producing updated state estimates and probabilistic forecasts. The arrows describe the modeled causal hypothesis; they do not establish molecular causality from the aggregate calibration data alone.

Biologically, the working hypothesis is that cells deposit and reorganize fibronectin, which supports integrin-mediated anchorage and persistent mesenchymal motion. Interstitial flow competes with this local matrix accumulation and organization. Reduced available fibronectin weakens effective cell–matrix adhesion and shifts the phenotype distribution toward rounded amoeboid motion, which is less adhesion dependent and, in the motivating experiment, less directionally persistent [12]. This is an effective coarse-grained hypothesis: the present model represents flow-mediated matrix loss by a spatially uniform kinetic term and does not solve fluid advection, matrix-fiber mechanics, integrin signaling, or proteolysis explicitly.

A conventional forward simulation cannot correct itself when a measured cell deviates from its predicted path. Moreover, microscopy usually reveals noisy positions while active velocity, adhesion state, and phenotype are incompletely observed. A digital twin addresses this mismatch by repeatedly predicting, receiving a measurement, updating hidden states and uncertainty, and forecasting again. This definition connects Kalman filtering [14, 15], sequential Monte Carlo [16–18], and biomedical digital-twin frameworks [19–21]. Here we ask whether a mechanistic model can reproduce phenotype-resolved observables, improve noisy positions online, identify phenotype from coordinates alone, and outperform simple forecast baselines. All trajectory-level tests use synthetic data; no validation on the original experimental tracks is claimed. Figure 1 summarizes the biological hypothesis and the prediction–measurement loop.

The primary claim is therefore deliberately bounded: a biologically structured mechanistic simulator constrained by published population observables can operate as an online probabilistic digital twin, modestly improve noisy synthetic state estimates, and produce calibrated prospective forecasts and testable biological predictions. We do not claim experimental trajectory-level validation, unique identification of molecular mechanism, or reliable phenotype classification from position alone.

## 2 Methods

### 2.1 Flow, matrix, adhesion, and phenotype

We use a deliberately low-dimensional biological hierarchy. Flow acts first on an effective extracellular-matrix variable, the matrix variable controls cell–matrix adhesion, adhesion shifts the stationary balance between motility phenotypes, and phenotype controls active motion. This hierarchy is intended to be interpretable and testable; it is not a molecularly complete signaling network.

The normalized fibronectin state  $f$  follows

$$\frac{df}{dt} = k_{\text{prod}}(1 - f) - k_{\text{loss}}f - k_{\text{sweep}}uf, \quad (1)$$

where  $u \in \{0, 1\}$  denotes no flow or flow. Parameters  $k_{\text{prod}} = 0.40$ ,  $k_{\text{loss}} = 0.10$ , and  $k_{\text{sweep}} = 1.0$  give  $f^*(0) = 0.80$  and  $f^*(1) = 0.267$ . Cell–matrix adhesion evolves as

$$\frac{da}{dt} = k_{\text{form}}f(1 - a) - k_{\text{off}}a, \quad (2)$$

with  $k_{\text{form}} = 0.60$  and  $k_{\text{off}} = 0.15$ , giving  $a^*(0) = 0.762$  and  $a^*(1) = 0.516$ .

The term  $k_{\text{sweep}}uf$  is an effective removal term. It represents the hypothesis that interstitial flow reduces locally available or stably assembled fibronectin, but it is not a Navier–Stokes velocity, a spatial advection operator, or a direct force on the cell. Because the available experimental constraints are steady population observables, trajectory simulations were initialized at the condition-specific equilibria  $f^*(u)$  and  $a^*(u)$ . Retaining Eqs. (1) and (2) makes the assumed causal mechanism explicit and permits later extension to time-dependent or spatially resolved matrix measurements.

Each cell has phenotype  $z_i \in \{A, M\}$ . The equilibrium mesenchymal probability is a logistic function of adhesion fixed by  $p_M^*(0) = 0.58$  and  $p_M^*(1) = 0.27$  [12]. Switching obeys

$$k_{AM} = \gamma p_M^*(a), \quad k_{MA} = \gamma[1 - p_M^*(a)], \quad (3)$$

with total rate  $\gamma = 0.25$  in inverse model-time units.

More explicitly,  $p_M^*(a) = [1 + \exp\{-\beta(a - a_c)\}]^{-1}$ , where  $\beta$  and  $a_c$  are determined by the two published phenotype fractions rather than independently fitted to trajectories. This construction holds the mean residence time scale,  $1/\gamma$ , fixed while allowing flow to change the equilibrium occupancy. A cell is classified as mesenchymal when  $z_i = M$  and amoeboid when  $z_i = A$ ; the state switches stochastically during a trajectory and is not inferred deterministically from instantaneous speed.

### 2.2 Active migration and pair interactions

Position and active velocity satisfy

$$d\mathbf{x}_i = \mathbf{v}_i dt + \mu \mathbf{F}_i dt + \sqrt{2D} d\mathbf{W}_i^{(x)}, \quad (4)$$

$$d\mathbf{v}_i = -\frac{\mathbf{v}_i}{\tau_{z_i}} dt + v_{z_i} \sqrt{\frac{2}{\tau_{z_i}}} d\mathbf{W}_i^{(v)}. \quad (5)$$

Here  $\mathbf{v}_i$  is the active velocity,  $D$  is translational diffusivity,  $\mu = 0.15$  is cell mobility, and  $\mathbf{W}_i^{(x)}$  and  $\mathbf{W}_i^{(v)}$  are independent two-dimensional Wiener processes. The net cell–cell force is  $\mathbf{F}_i = \sum_{j \neq i} \mathbf{F}_{ij}$ .

Table 1: Frozen Version 2 model and observation parameters. All dimensional quantities are in model units.

| Symbol           | Value      | Interpretation                        |
|------------------|------------|---------------------------------------|
| $\Delta t$       | 0.01       | fine integration step                 |
| $D$              | 1.20       | translational diffusivity             |
| $\tau_A, \tau_M$ | 0.70, 18.0 | active-velocity persistence times     |
| $v_A, v_M$       | 2.30, 1.00 | active-velocity noise scales          |
| $\gamma$         | 0.25       | total phenotype-switching rate        |
| $\mu$            | 0.15       | mobility multiplying pair force       |
| $d, R$           | 1.0, 1.5   | cell diameter and force cutoff        |
| $k_r, k_a$       | 300, 20    | repulsive and adhesive stiffness      |
| $\sigma_y$       | 0.50       | localization-error standard deviation |
| $N_e$            | 100        | digital-twin ensemble size            |

With  $\mathbf{r}_{ij} = \mathbf{x}_i - \mathbf{x}_j$ ,  $r_{ij} = \|\mathbf{r}_{ij}\|$ , and  $\hat{\mathbf{r}}_{ij} = \mathbf{r}_{ij}/r_{ij}$ , the implemented pair force is

$$\mathbf{F}_{ij} = \begin{cases} k_r(d - r_{ij})\hat{\mathbf{r}}_{ij}, & 0 < r_{ij} < d, \\ -k_a q_{z_i z_j} \frac{(r_{ij} - d)(R - r_{ij})}{R - d} \hat{\mathbf{r}}_{ij}, & d \leq r_{ij} < R, \\ \mathbf{0}, & r_{ij} \geq R, \end{cases} \quad (6)$$

where  $d = 1$  is the effective cell diameter,  $R = 1.5d$  is the adhesive cutoff,  $k_r = 300$  is the repulsive stiffness, and  $k_a = 20$  is the adhesive stiffness. The phenotype factors are  $q_{MM} = 1.0$ ,  $q_{AM} = q_{MA} = 0.6$ , and  $q_{AA} = 0.2$ . Because  $\mathbf{F}_{ji} = -\mathbf{F}_{ij}$ , internal pair forces sum to zero. A hard-core projection removed any residual numerical overlap after each time step. The frozen motility parameters were  $D = 1.20$ ,  $\tau_A = 0.70$ ,  $\tau_M = 18.0$ ,  $v_A = 2.30$ , and  $v_M = 1.00$ . Pair-force parameters were not calibrated to measured colony geometry and are treated as a sensitivity component rather than a validated force law.

Equation (6) is a phenomenological finite-range contact law used in the present numerical implementation. It is not the Gaussian pair potential used in the earlier analytical theory [13], where Gaussian potentials made the field-theoretic treatment tractable. Both descriptions represent short-range repulsion and adhesion, but they are not mathematically identical. Establishing quantitative continuity with that theory would require implementing its Gaussian potential and repeating the interacting-colony and digital-twin validation.

The active-velocity equation is an Ornstein–Uhlenbeck process. Conditional on a fixed phenotype, its autocorrelation decays on time scale  $\tau_{z_i}$ , providing an explicit colored active-noise contribution in addition to the independent translational fluctuation in Eq. (4). Thus  $v_A$  and  $v_M$  set active-velocity scales, whereas  $D$  controls unresolved positional fluctuations. The numerical time step was  $\Delta t = 0.01$ , total duration was 50 model-time units, and no direct conversion to physical hours is assumed.

### 2.3 Calibration and held-out evaluation

Experimental targets were phenotype resolved: mesenchymal and amoeboid directional persistence  $0.21 \pm 0.02$  and  $0.13 \pm 0.02$ , MSD exponents 1.46 and 1.27, and approximate speed ratio 1.09 [12]. Persistence was net displacement divided by path length. Candidate parameters were screened on development seeds, three finalists were compared on ten independent validation seeds, and the

Table 2: Parameter provenance. “Calibrated” denotes selection using development simulations before the final held-out seeds; it does not imply direct estimation from raw experimental trajectories.

| Quantity                             | Status                    | Basis                                                            |
|--------------------------------------|---------------------------|------------------------------------------------------------------|
| $p_M^*(0), p_M^*(1)$                 | Published targets         | Experimental endpoint fractions [12]                             |
| Persistence, MSD, speed ratio        | Published targets         | Phenotype-resolved aggregate observables [12]                    |
| $D, \tau_A, \tau_M, v_A/v_M$         | Calibrated                | Development grids and independent finalist selection             |
| Matrix kinetic rates                 | Modeling assumptions      | Chosen to encode the hypothesized flow-matrix pathway            |
| Pair-force parameters                | Modeling assumptions      | Stable short-range repulsion/adhesion; not fitted to colony data |
| $\sigma_y, N_e$ and likelihood scale | Observation/twin settings | Fixed before final multi-experiment evaluation                   |
| Random seeds                         | Reproducibility controls  | Disjoint development, selection, and final-test seed ranges      |

selected parameters were frozen. Final evaluation used 20 new seeds, yielding 257 dominantly mesenchymal and 383 dominantly amoeboid trajectories. Final-test results were not used for retuning.

Trajectories were sampled at 192 approximately equally spaced frames. For a sampled trajectory  $\{\mathbf{x}_{i,n}\}_{n=0}^{N-1}$ , directional persistence was

$$P_i = \frac{\|\mathbf{x}_{i,N-1} - \mathbf{x}_{i,0}\|}{\sum_{n=0}^{N-2} \|\mathbf{x}_{i,n+1} - \mathbf{x}_{i,n}\|}. \quad (7)$$

A trajectory was called phenotype dominant when the corresponding phenotype occupied at least 50% of its sampled frames. This trajectory-level label is used only for aggregate calibration and is distinct from the frame-level phenotype inferred by the digital twin.

For each phenotype-dominant group, the time-averaged ensemble MSD was

$$\overline{\delta^2}(\ell) = \frac{1}{(N-\ell)N_g} \sum_{i=1}^{N_g} \sum_{n=0}^{N-\ell-1} \|\mathbf{x}_{i,n+\ell} - \mathbf{x}_{i,n}\|^2, \quad (8)$$

where  $N_g$  is the number of trajectories in the group. The exponent  $\alpha$  was obtained from an ordinary least-squares fit of  $\log \overline{\delta^2}$  versus  $\log \Delta$  over the prespecified lag interval  $5 \leq \Delta \leq 25$ , so that  $\overline{\delta^2} \propto \Delta^\alpha$ . The fitting window was fixed before the held-out evaluation and was not selected separately for the two phenotypes.

Parameter development proceeded in three stages. A coarse grid established a region compatible with population observables; a focused search varied  $D$ ,  $\tau_A$ ,  $\tau_M$ , and  $v_A$ ; and three finalists were compared on ten independent seeds. The selected tuple  $(D, \tau_A, \tau_M, v_A) = (1.20, 0.70, 18.0, 2.30)$  was then frozen before evaluation on 20 new seeds. This separation reduces, but does not eliminate, overfitting to the small set of aggregate experimental targets.

## 2.4 Synthetic microscopy and hybrid update

Full interacting simulations generated hidden truth for 16 cells. Positions were observed every 1.0 model-time unit with

$$\mathbf{y}_{i,k} = \mathbf{x}_{i,k} + \boldsymbol{\epsilon}_{i,k}, \quad \boldsymbol{\epsilon}_{i,k} \sim \mathcal{N}(\mathbf{0}, 0.5^2 \mathbf{I}). \quad (9)$$

The twin used 100 members. Continuous states  $[x, y, v_x, v_y]$  were updated locally using an ensemble Kalman update. Phenotype was updated with position-likelihood weights of standard deviation  $2.5\sigma_y$  and systematic resampling.

For each cell, let  $\mathbf{s}_{i,k} = [x, y, v_x, v_y]^T$  and let  $H$  select its two position components. From the forecast ensemble we computed the sample cross-covariance  $P_{sy}$  and predicted-observation covariance  $P_{yy}$ . The local gain was

$$K = P_{sy}(P_{yy} + \sigma_y^2 \mathbf{I})^{-1}, \quad (10)$$

and each member was updated with a perturbed observation. Updating cells locally avoids estimating a poorly conditioned full  $64 \times 64$  covariance from only 100 members.

The discrete phenotype was not averaged through Eq. (10). Instead, member  $j$  for cell  $i$  received likelihood

$$w_{i,k}^{(j)} \propto \exp \left[ -\frac{\|\mathbf{y}_{i,k} - \mathbf{x}_{i,k}^{(j)}\|^2}{2(2.5\sigma_y)^2} \right], \quad (11)$$

followed by normalization and systematic resampling. The posterior mesenchymal probability was the weighted ensemble fraction in state  $M$ . The broader phenotype likelihood acknowledges that one noisy displacement is weak evidence for a motility state.

The operational cycle was therefore: (i) propagate every ensemble member through the mechanistic equations; (ii) observe noisy positions; (iii) update continuous position and velocity; (iv) reweight and resample phenotype; and (v) use the posterior ensemble as the initial condition for the next prediction or a measurement-free forecast. Hidden truth was accessed only for retrospective scoring.

An ablation showed that removing the pair force changed final radius by only  $-0.025$  without flow and  $-0.029$  with flow; both 95% confidence intervals included zero. Efficient multi-experiment filtering therefore used a reduced predictor without explicit pair-force evaluation, while hidden truth retained full interacting dynamics and hard-core exclusion. The reduced predictor was 797-fold faster over one observation interval, changed ensemble-mean position by only 0.031 (below localization noise), and preserved phenotype evolution exactly.

Ten independent experiments of 16 cells and 50 updates were used for assimilation tests. We report Euclidean RMSE, nominal-95% coordinate coverage, phenotype ROC AUC, balanced accuracy, and Brier score. Ten additional experiments were assimilated through a cutoff and propagated without future measurements to horizons 0.5, 1, 2, 4, and 8. Baselines held the last position or linearly extrapolated recent motion.

Position RMSE is the square root of the mean squared Euclidean error across cells and frames. Coordinate coverage is the fraction of true  $x$  and  $y$  coordinates lying between the 2.5th and 97.5th ensemble percentiles. The Brier score is the mean squared error between posterior mesenchymal probability and the binary hidden state. Forecast comparisons were paired by experiment. We report mean differences, 95% confidence intervals based on the paired sample, and two-sided paired  $t$  tests as descriptive measures; no correction for multiple horizons was applied.

Table 3: Final held-out mechanistic validation across 20 independent replicates. Values are replicate mean  $\pm$  SD. Experimental values are aggregate targets from Ref. [12].

| Observable                 | Simulation        | Target | Relative difference |
|----------------------------|-------------------|--------|---------------------|
| Mesenchymal persistence    | $0.217 \pm 0.036$ | 0.210  | +3.3%               |
| Amoeboid persistence       | $0.151 \pm 0.019$ | 0.130  | +16.2%              |
| Mesenchymal MSD exponent   | $1.444 \pm 0.122$ | 1.460  | -1.1%               |
| Amoeboid MSD exponent      | $1.212 \pm 0.101$ | 1.270  | -4.6%               |
| Amoeboid/mesenchymal speed | $1.045 \pm 0.021$ | 1.090  | -4.1%               |

## 2.5 Levels of evidence

Four levels of evidence are kept separate. First, *biological calibration* compares simulated aggregate observables with published targets. Second, *synthetic internal validation* tests state reconstruction and uncertainty against hidden simulated truth generated from recorded seeds. Third, *prospective model prediction* compares flow and no-flow colony simulations after parameters are frozen. Fourth, *future experimental validation* would require raw microscopy tracks, synchronized morphology or molecular phenotype labels, and matrix measurements not available in the present study. This hierarchy prevents synthetic ground-truth performance from being presented as experimental validation.

## 3 Results

### 3.1 Mechanistic validation

Held-out mesenchymal persistence was  $0.217 \pm 0.036$  versus 0.210; amoeboid persistence  $0.151 \pm 0.019$  versus 0.130; mesenchymal MSD exponent  $1.444 \pm 0.122$  versus 1.460; amoeboid exponent  $1.212 \pm 0.101$  versus 1.270; and speed ratio  $1.045 \pm 0.021$  versus 1.090. Four relative differences were within 5%; amoeboid persistence was 16.2% high. Late mesenchymal fractions under flow and no flow were  $0.308 \pm 0.074$  and  $0.583 \pm 0.085$ .

The MSD curves remained superdiffusive over the fitting interval for both phenotype-dominant groups (Fig. 2). The larger mesenchymal exponent is consistent with the longer active-velocity correlation time assigned to that state. The purpose of the exponent comparison is not to claim a unique stochastic mechanism from MSD alone, but to verify that the same model producing phenotype switching also reproduces the observed ordering and approximate magnitude of anomalous transport.

### 3.2 State estimation and phenotype identifiability

Across ten experiments, open-loop predictions had RMSE  $3.312 \pm 0.061$ . Raw measurements had RMSE  $0.700 \pm 0.009$ , and posterior estimates achieved  $0.689 \pm 0.008$ . Relative improvement was  $1.63 \pm 0.61\%$ ; all ten experiments improved. The paired reduction was 0.0114 (95% CI 0.0083–0.0145;  $p = 1.57 \times 10^{-5}$ ), and 327 of 500 frames improved. Coverage was  $0.934 \pm 0.006$ , with interval width  $1.841 \pm 0.004$ .

Phenotype inference was weak: AUC  $0.565 \pm 0.050$ , balanced accuracy  $0.524 \pm 0.031$ , and Brier score  $0.209 \pm 0.020$ . In the development experiment, the twin Brier score (0.227) did not beat constant equilibrium probability (0.226) or observed fraction (0.222). Positions therefore contained weak ranking information but did not support reliable frame-level phenotype reconstruction.

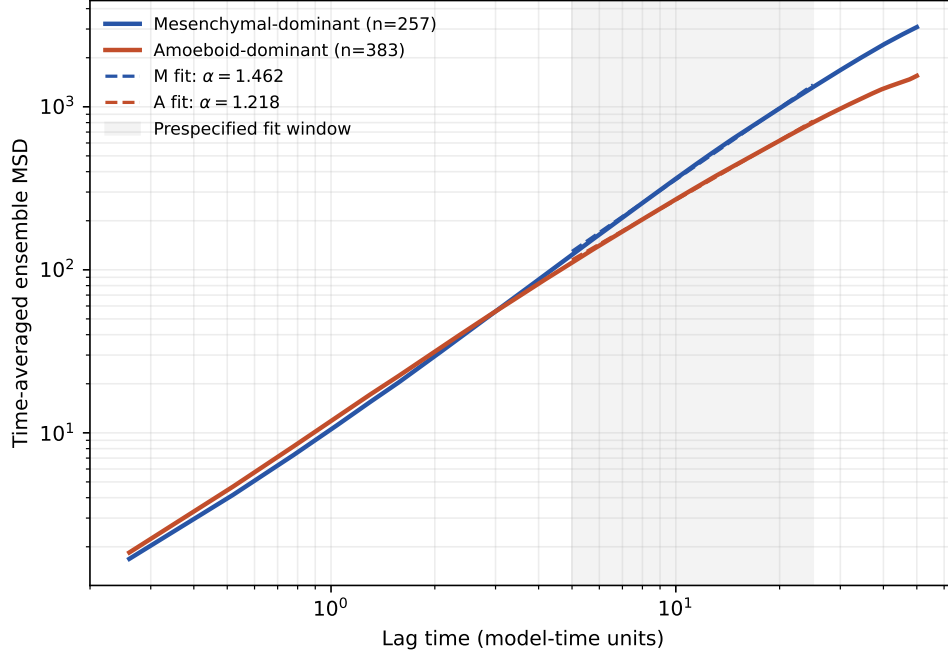


Figure 2: Held-out phenotype-resolved mean-squared displacement. Solid curves are time-averaged ensemble MSDs pooled from the 20 final seeds under both flow conditions and grouped by dominant phenotype. Dashed lines are power-law fits over the prespecified shaded interval  $5 \leq \Delta \leq 25$ . The slopes are reported in the legend.

This contrast is informative. The continuous update produced a statistically detectable but practically modest denoising benefit, whereas the discrete hidden state was poorly identified. High positional coverage did not imply accurate phenotype recovery: the position ensemble could encompass the truth while different phenotype histories generated overlapping short-time displacements. We therefore treat phenotype probability as an uncertainty-bearing latent variable, not as a reliable microscopy label.

Figure 3 illustrates the operational behavior for cell 1 of the fixed development experiment (truth seed 80000); the cell was specified by index rather than selected by performance. Measurements repeatedly pull the prior trajectory toward the hidden path, whereas the posterior phenotype probability frequently remains intermediate or disagrees with the binary hidden state. After the final update, the measurement-free forecast broadens rapidly. The hidden future is usually compatible with the widening coordinate intervals even when the ensemble mean misses its detailed path. This single realization is explanatory only; all quantitative claims use the multi-experiment summaries.

### 3.3 Prospective forecasting

The twin outperformed linear extrapolation at every horizon by 9.9–48.7%; paired tests were significant throughout. Its advantage over holding position was smaller (3.7–5.2%): significant at 0.5, 1, and 8 units and positive but not conventionally significant at 2 and 4 units. Mean nominal-95% coverage was 0.934.

The increasing advantage over linear extrapolation has a mechanistic interpretation. Linear continuation assumes indefinitely persistent velocity, whereas both Ornstein–Uhlenbeck decorrelation and stochastic phenotype switching bend or redirect future paths. The hold baseline is more

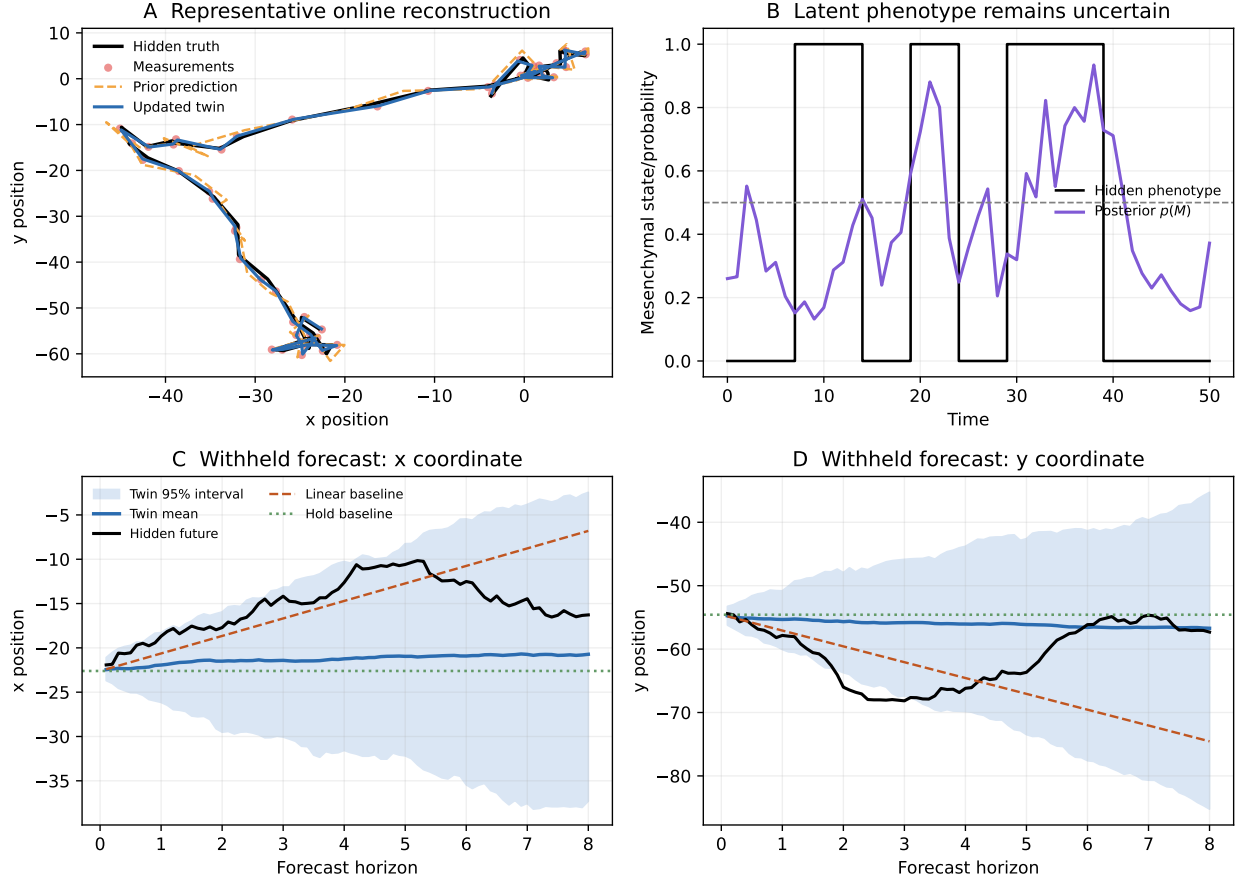


Figure 3: Representative Version 2 digital-twin realization for the prespecified first cell. (A) Hidden trajectory, noisy measurements, prior means, and posterior means through 50 measurement updates. (B) Hidden binary phenotype and posterior mesenchymal probability. Imperfect agreement is consistent with weak aggregate phenotype identifiability. (C,D) Measurement-free forecasts for the  $x$  and  $y$  coordinates after the final update. Shading gives coordinate-wise 95% ensemble intervals; hold-position and linear baselines are shown for context. This fixed-seed illustration is not used for aggregate performance claims.

conservative and consequently harder to beat at long horizon. Forecast intervals widened from a mean coordinate width of 5.77 at horizon 0.5 to 38.27 at horizon 8, while coverage stayed between 0.928 and 0.938. The twin therefore expressed decreasing predictability through wider distributions rather than an unjustified point prediction.

### 3.4 Prospective colony prediction

In 20 paired simulations, flow reduced radius from  $52.26 \pm 7.47$  to  $42.67 \pm 5.80$  (paired change  $-9.59$ , 95% CI  $-12.81$  to  $-6.37$ ,  $p = 5.51 \times 10^{-6}$ ). Nearest-neighbor distance decreased from  $22.30 \pm 4.03$  to  $17.26 \pm 3.94$ , and late mesenchymal fraction decreased by 47.1%. Removing the pair force left the radius effect unchanged ( $-12.486$  with force;  $-12.482$  without), indicating that compaction arose from the flow-matrix-phenotype-persistence pathway rather than weak short-range adhesion.

The term “compaction” here is relative: cells still dispersed from their initial configuration, but

Table 4: Withheld forecast validation (ten experiments, 16 cells each).

| Horizon | Twin   | Hold   | Linear | vs. hold | vs. linear | Coverage |
|---------|--------|--------|--------|----------|------------|----------|
| 0.5     | 2.139  | 2.238  | 2.374  | 4.4%     | 9.9%       | 0.938    |
| 1.0     | 3.376  | 3.560  | 4.081  | 5.2%     | 17.3%      | 0.928    |
| 2.0     | 5.423  | 5.643  | 7.485  | 3.9%     | 27.6%      | 0.934    |
| 4.0     | 8.926  | 9.272  | 14.546 | 3.7%     | 38.6%      | 0.934    |
| 8.0     | 14.212 | 14.784 | 27.703 | 3.9%     | 48.7%      | 0.938    |

the flow-conditioned population occupied a smaller final radius and had a smaller nearest-neighbor distance than its paired no-flow counterpart. Because direct pair-force ablation barely changed radius, this prediction should not be interpreted as formation of a strongly cohesive aggregate. Rather, flow shifts occupancy toward the shorter-persistence state, decreasing persistence-driven spreading over the simulated duration.

## 4 Discussion

The main advance is mechanistic rather than a larger denoising score. Version 2 makes the causal structure explicit: flow removes matrix material, the matrix controls adhesion, adhesion sets switching equilibrium, phenotype determines active speed and persistence, and stochastic dynamics generate trajectories. Calibration, model selection, and held-out testing were separated by seed.

This causal chain gives biological meaning to the twin’s latent variables. Fibronectin and adhesion represent an effective microenvironmental memory, phenotype represents motility plasticity, and active velocity represents persistent polarity that cannot be inferred from a single image. The variables are therefore useful not merely as curve-fitting coordinates but as experimentally addressable hypotheses: matrix staining or adhesion reporters could constrain  $f$  and  $a$ , morphology or molecular markers could constrain  $z_i$ , and denser imaging could constrain  $v_i$ .

The modest but consistent RMSE reduction shows that the dynamical prior adds information beyond accurate localization measurements. Forecasting is the stronger operational result. Linear extrapolation becomes unrealistic as velocity decorrelates and phenotype switches, so the twin’s advantage grows with horizon. Approximately nominal coverage communicates increasing uncertainty rather than hiding it.

Version 2 should not be read as a numerical upgrade chosen to maximize every metric over the simpler Version 1 model. The two analyses use different mechanistic assumptions and validation cohorts. The more explicit matrix and interaction hierarchy reduces the synthetic assimilation gain from 14.5% in Version 1 to 1.63% here, while preserving a substantial advantage over linear extrapolation. This is a scientifically useful tradeoff: added biological structure exposes model mismatch and identifiability limitations that a simpler data-generating model can conceal. Accordingly, we present Version 1 as a proof of operational feasibility and Version 2 as the more stringent mechanistic test, not as directly ranked competitors.

The phenotype result is equally important. Coordinates alone yielded weak discrimination and no Brier-score advantage over constant probability. A twin should reveal when a hidden variable is not identifiable, not force a confident label. Morphology, adhesion reporters, matrix imaging, or molecular markers are needed to update phenotype directly.

The colony calculation predicts that flow produces a more compact distribution by reducing long-persistence mesenchymal occupancy. The ablation shows that short-range adhesion is present

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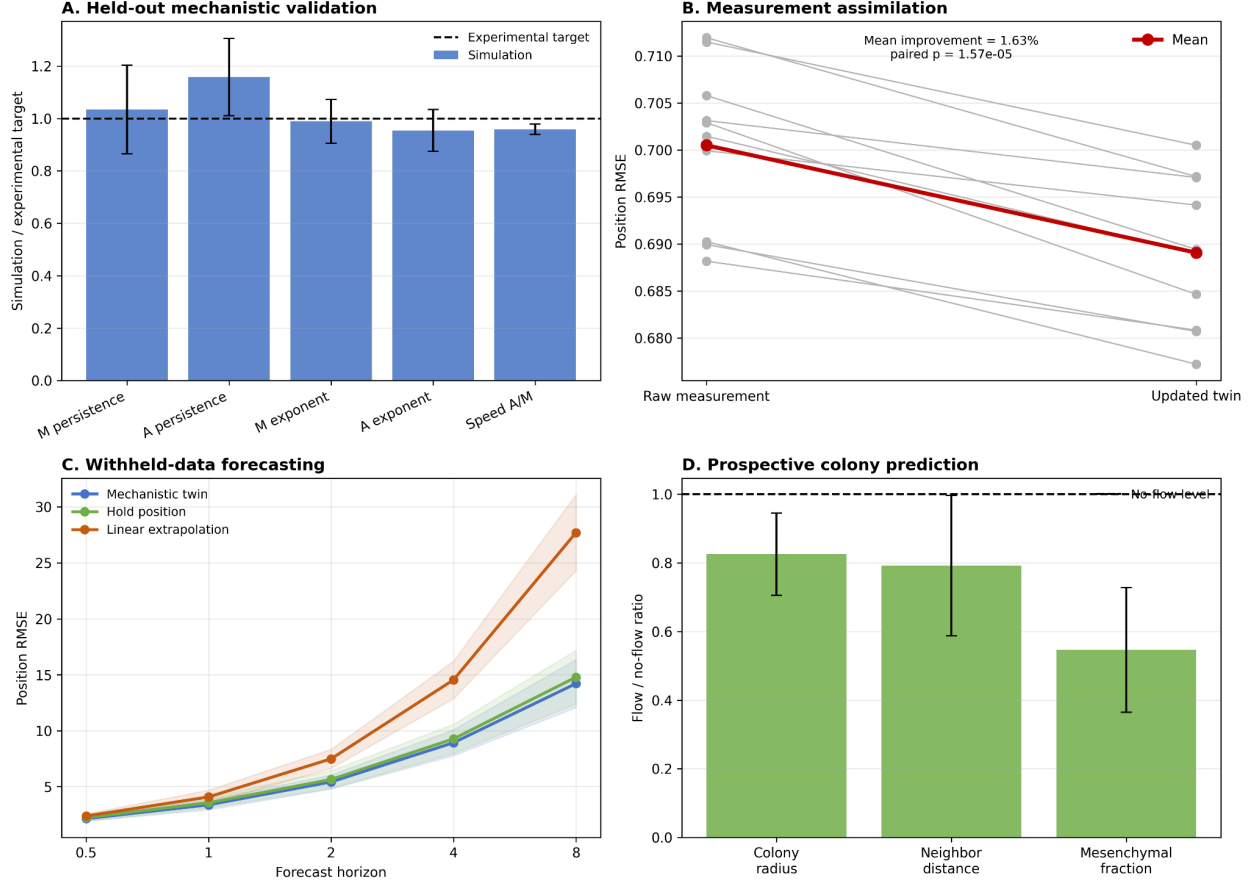


Figure 4: Version 2 summary. (A) Held-out phenotype-resolved validation. (B) Paired measurement and posterior RMSE. (C) Withheld forecast RMSE (mean  $\pm$  SD). (D) Prospective flow/no-flow ratios for colony observables; this panel is a model prediction, not experimental validation of colony geometry.

but negligible after rapid dispersal in this unconstrained geometry. We do not claim a cohesive spheroid or attribute compaction to cell-cell adhesion.

Several predictions are experimentally falsifiable. First, under matched observation time, flow-conditioned populations should display reduced long-lag spreading after controlling for localization error. Second, synchronized matrix and morphology imaging should improve phenotype inference beyond the weak AUC obtained from coordinates alone. Third, if direct cell-cell adhesion is important in a denser or confined assay, removing the pair force should materially change colony geometry; the negligible ablation effect observed here predicts that it will not do so in the present dilute regime. These tests distinguish the causal components of the model rather than merely checking another aggregate fit.

Limitations are substantial. Aggregate observables rather than raw trajectories constrained the model. Filtering and forecasting are synthetic and can be optimistic under model misspecification. Parameters are effective and in model units. Matrix states are spatially uniform; the sweep term is an effective loss mechanism rather than a solved interstitial-flow field. Direct cell advection, spatial

Table 5: Scope of the present model. The right column identifies abstractions that should not be inferred from the implemented equations.

| Process               | Present representation                | Not represented                                    |
|-----------------------|---------------------------------------|----------------------------------------------------|
| Interstitial flow     | Binary condition and matrix-loss rate | Spatial fluid velocity or cell advection           |
| Fibronectin           | Uniform scalar state                  | Fibers, transport, assembly, and deformation       |
| Cell–matrix adhesion  | Uniform scalar state                  | Receptor-level binding and focal adhesions         |
| Phenotype plasticity  | Two-state Markov switching            | Continuous morphology and signaling networks       |
| Active migration      | Phenotype-specific colored noise      | Explicit polarity and cytoskeletal mechanics       |
| Cell–cell interaction | Finite-range pair force and hard core | Junction dynamics and tissue rheology              |
| Microscopy            | Gaussian position error               | Segmentation errors, missed cells, and track swaps |
| Digital twin          | Ensemble state update and forecast    | Patient-specific or clinical validation            |

fibronectin transport and assembly, matrix deformation, receptor signaling, and proteolysis are absent. The implemented contact force is a short-range numerical law rather than the Gaussian interaction of the earlier theory, and its ablation shows that pair interactions are minimal in the present dilute geometry. The results demonstrate internal predictive consistency, not experimental or clinical validation. Blinded testing on real tracks with synchronized morphology and matrix measurements is the decisive next step.

A global sensitivity or structural-identifiability analysis was not performed. Development grids established that multiple parameter combinations can approximate the aggregate targets, which is another reason to interpret the frozen parameters as an effective model rather than unique biological constants. A stronger robustness study should vary matrix-loss and switching parameters, repeat the final tests with the Gaussian potential of Ref. [13], and evaluate whether forecast rankings and the flow-induced colony prediction persist.

## 5 Conclusion

We developed a matrix- and phenotype-aware migration digital twin that reproduces phenotype-resolved motility statistics, modestly but consistently improves noisy synthetic positions, and outperforms linear extrapolation in withheld forecasts with near-nominal coverage. It also yields two falsifiable conclusions: flow-mediated phenotype redistribution should reduce persistence-driven population dispersal, and coordinates alone should be insufficient for reliable instantaneous phenotype identification.

## Code and data availability

The notebook, parameters, seeds, and generated outputs are available at <https://github.com/himadriut/interstitial-flow-digital-twin>. No original experimental trajectories are included; all trajectory-level validation data are synthetic.

## Author contributions

H.S.S. conceived the study, developed the model and computational experiments, analyzed the results, and wrote the manuscript.

## Competing interests

The author declares no competing interests.

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